LeadForge AI — Complete Architecture & Master Prompt Document

LeadForge AI is a fully free and open-source AI-powered lead generation and digital presence analysis platform. This document contains: - Product overview - System architecture - Agent architecture - Tech stack - Development roadmap - Complete architecture flow - Full master prompt

Project Vision

```
Build an AI-powered SaaS platform that helps agencies, freelancers, and marketers identify businesses with poor
```

Core Features

```
1. Business Lead Search
2. Digital Presence Analysis
3. SEO Scoring Engine
4. Social Media Detection
5. AI Outreach Generator
6. CSV/Excel Export
7. Analytics Dashboard
8. Saved Leads Management
```

Recommended Tech Stack

```
Frontend:
- React
- Tailwind CSS
- Vite
- Shadcn UI
- Framer Motion
- Recharts

Backend:
- Python FastAPI

Scraping:
- Playwright
- BeautifulSoup
- Requests

AI:
- Ollama
- Llama3 / Mistral / Gemma

SEO:
- Google Lighthouse

Database:
- SQLite

Automation:
- CrewAI
- n8n
```

System Architecture Flow

```
USER SEARCH
    ↓
```

```
FastAPI Backend
    ↓
Scraper Agent
(Playwright + BeautifulSoup)
    ↓
Business Data Extraction
    ↓
Website Analyzer
    ↓
Lighthouse SEO Analysis
    ↓
Digital Score Generator
    ↓
Ollama AI Outreach Agent
    ↓
Dashboard Results
    ↓
CSV Export
```

Agent Architecture

```
1. Scraper Agent
- Finds businesses
- Extracts websites
- Collects emails/phones

2. Analyzer Agent
- Checks responsiveness
- Runs Lighthouse
- Detects SEO issues

3. Scoring Agent
- Generates digital score
- Creates weakness reports

4. Outreach Agent
- Generates cold emails
- Generates LinkedIn DMs
- Generates SEO pitches
```

Database Schema

```
LEADS TABLE:
- id
- company_name
- website
- email
- phone
- score
- weaknesses
- outreach_message
- created_at

SEARCH_HISTORY TABLE:
- id
- query
- total_results
- created_at
```

Frontend Architecture

```
frontend/
  src/
    components/
    pages/
    services/
    hooks/
```

```
    layouts/
    charts/
    tables/
```

Backend Architecture

```
backend/
  api/
  services/
  scraper/
  seo/
  ai/
  database/
  models/
  utils/
```

Development Roadmap

```
WEEK 1:
- Landing page
- Dashboard UI
- FastAPI setup
- Lead search

WEEK 2:
- Scraping engine
- SEO analyzer
- Lighthouse integration

WEEK 3:
- AI outreach
- CSV export
- Saved leads

WEEK 4:
- Analytics
- Automation workflows
- Deployment
```

Complete Master Prompt

```
You are a senior AI engineer, SaaS architect, and full-stack developer.

Build a FULLY FREE and OPEN-SOURCE AI-powered Lead Generation and Digital Presence Analyzer platform.

IMPORTANT:
The project MUST use ONLY free and open-source tools.
DO NOT use paid APIs or proprietary services.
DO NOT depend on OpenAI paid APIs.

The platform should work completely locally or with free tiers.

PROJECT GOAL:
Build a modern SaaS-style web application that:
1. Finds businesses
2. Analyzes their digital presence
3. Scores their online quality
4. Generates AI outreach messages
5. Exports leads

TECH STACK:
Frontend:
- React
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```

- Recharts

Backend:
- Python FastAPI

Database:
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Scraping:
- Playwright
- BeautifulSoup
- Requests

AI:
- Ollama
- Llama3
- Mistral

SEO:
- Google Lighthouse

FEATURES:
- Landing Page
- Dashboard
- Lead Search
- SEO Analyzer
- Social Presence Detector
- AI Outreach Generator
- CSV Export
- Saved Leads

LANDING PAGE:
Create a premium dark-themed SaaS landing page with:
- Hero section
- Animated backgrounds
- CTA
- Testimonials placeholder
- Pricing placeholder

DASHBOARD:
Create:
- Sidebar navigation
- Charts
- Lead tables
- Analytics cards

LEAD SEARCH:
Allow searching:
- Business niche
- City
- Keywords

Example:
'Real estate companies in Hyderabad'

DIGITAL ANALYZER:
Check:
- Website existence
- HTTPS
- Mobile responsiveness
- SEO tags
- Social links
- Performance

LIGHTHOUSE:
Integrate Lighthouse for:
- SEO score
- Accessibility
- Performance

AI OUTREACH:
Generate:
- Cold emails
- LinkedIn DMs
- Website redesign pitches

OUTPUT:
Generate:
- Full folder structure

- Frontend code
- Backend code
- APIs
- Setup instructions
- Deployment guide

IMPORTANT:
The application must actually work.
Use scalable and clean architecture.